Chapter 36

Resource Acquisition and Transport in Vascular Plants

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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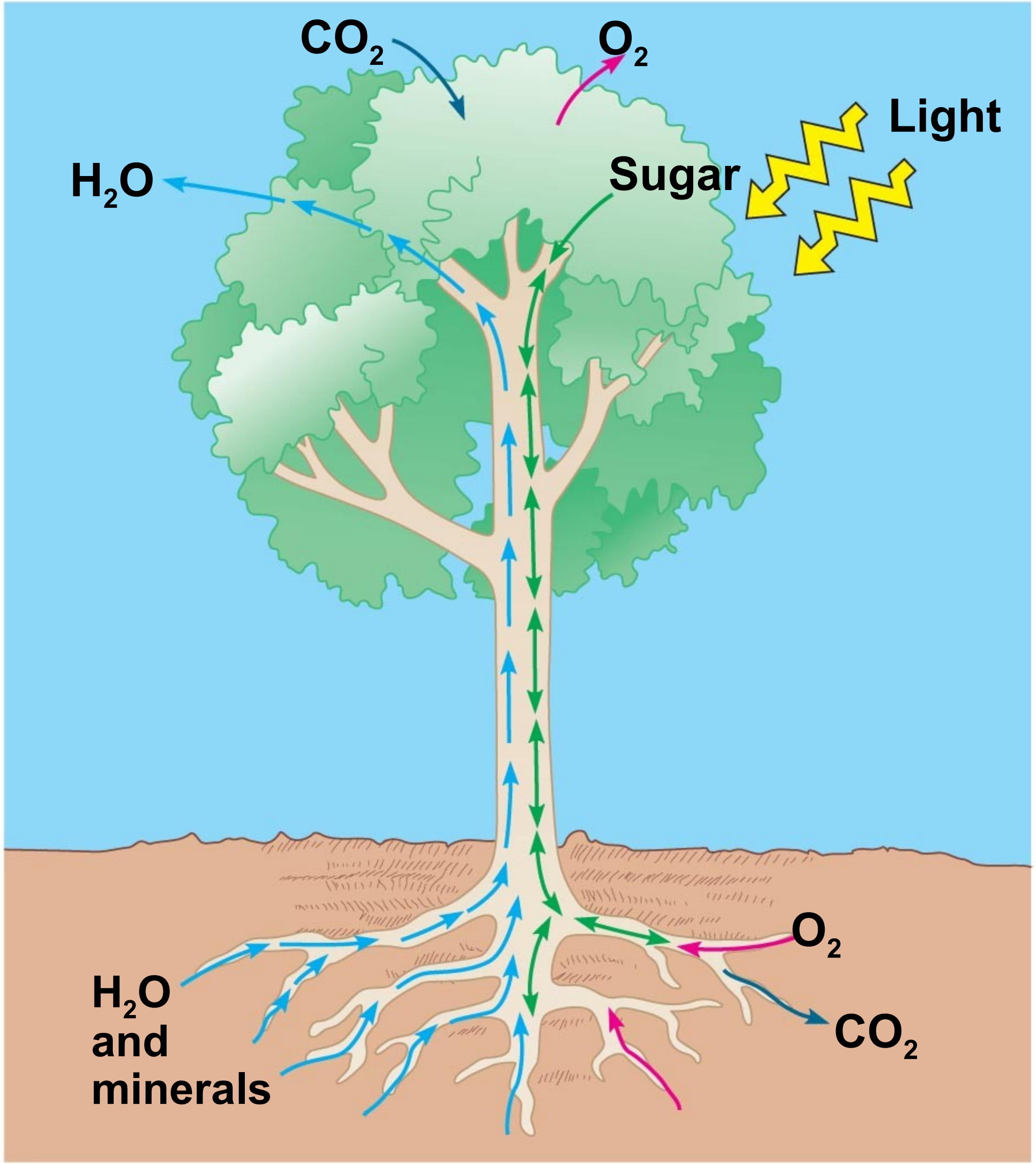
Overview: Underground Plants

- The success of plants depends on their ability to gather and conserve resources from their environment.
- The transport of materials is central to the integrated functioning of the whole plant.
- Diffusion, active transport, and bulk flow work together to transfer water, minerals, and sugars.

Resource Acquisition and Transport



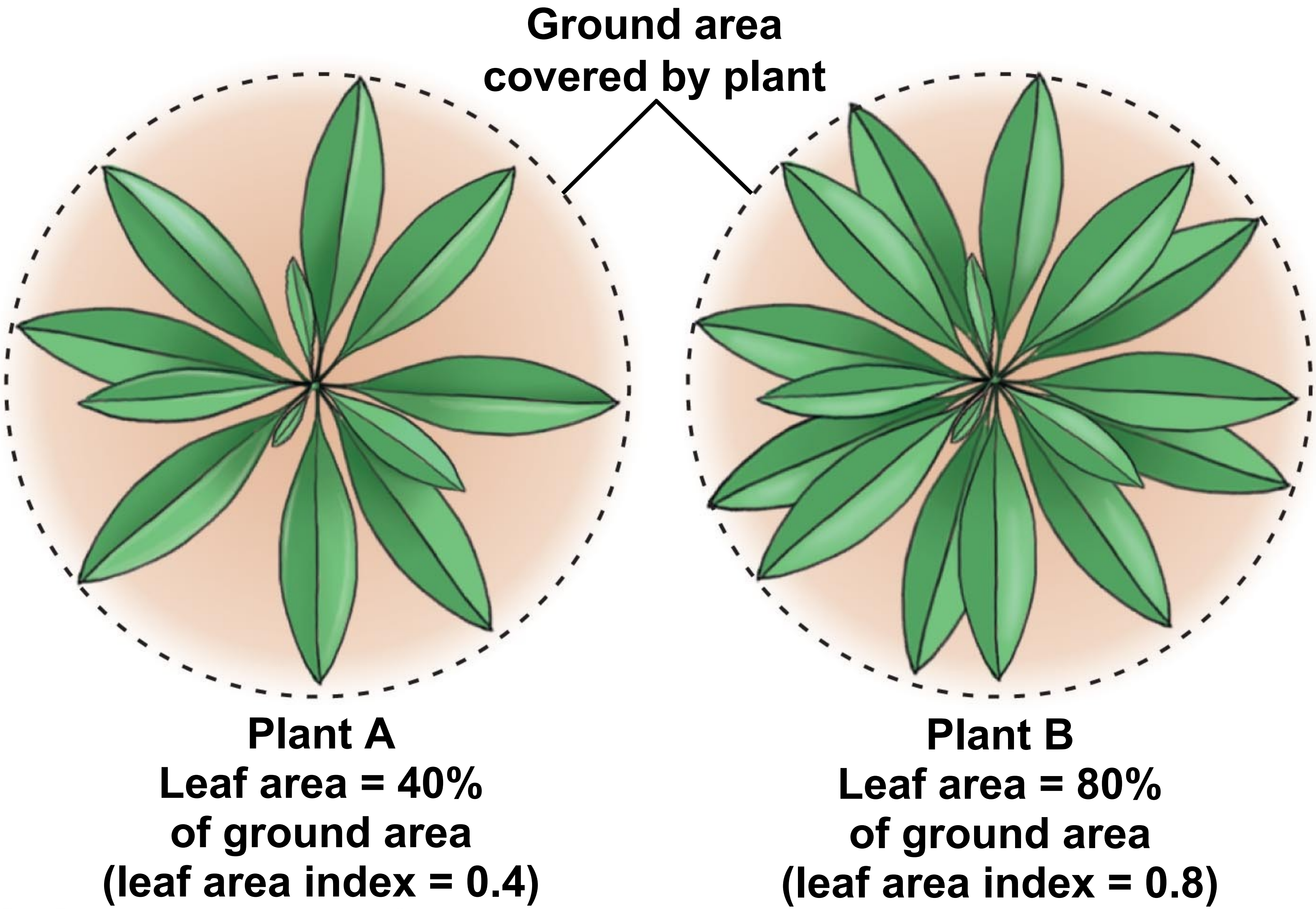
Concept 36.1: Land plants acquire resources both above and below ground

- The algal ancestors of land plants absorbed water, minerals, and CO_2 directly from the surrounding water.
- The evolution of xylem and phloem in land plants made possible the long-distance transport of water, minerals, and products of photosynthesis.
- ***Adaptations in each species represent compromises between enhancing photosynthesis and minimizing water loss***.

Shoot Architecture and Light Capture

- Stems serve as conduits for water and nutrients, and as supporting structures for leaves.
- **Phyllotaxy**, the arrangement of leaves on a stem, is specific to each species.
- Light absorption is affected by the ***leaf area index***, the ratio of total upper leaf surface of a plant divided by the surface area of land on which it grows.
- Leaf orientation affects light absorption.

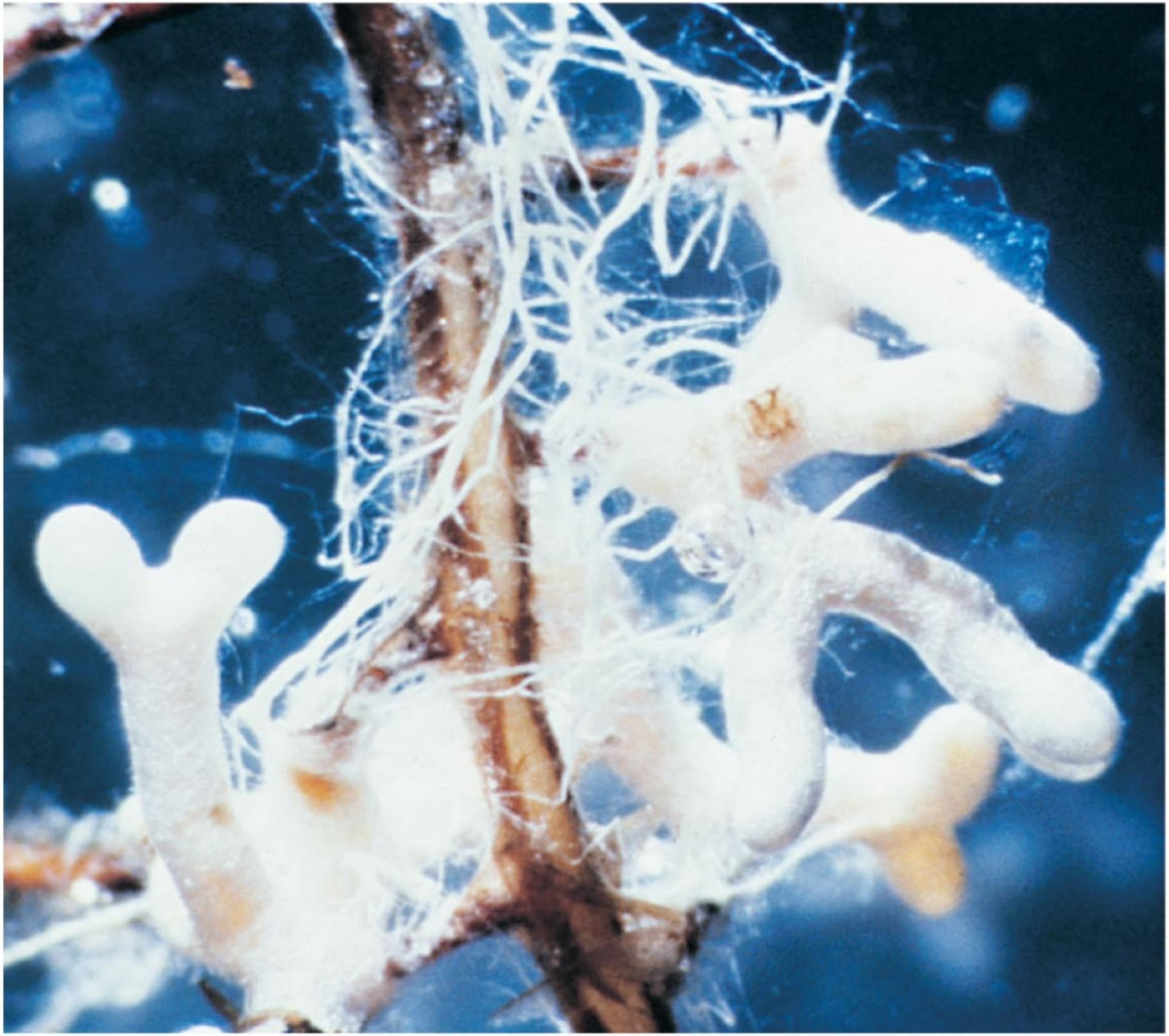
Leaf area index



Root Architecture and Acquisition of Water and Minerals

- Soil is a resource mined by the root system.
- Taproot systems anchor plants and are characteristic of most trees.
- Roots and the hyphae of soil fungi form symbiotic associations called **mycorrhizae**.
- Mutualisms with fungi helped plants colonize land.

mycorrhiza, a symbiotic association of fungi and roots



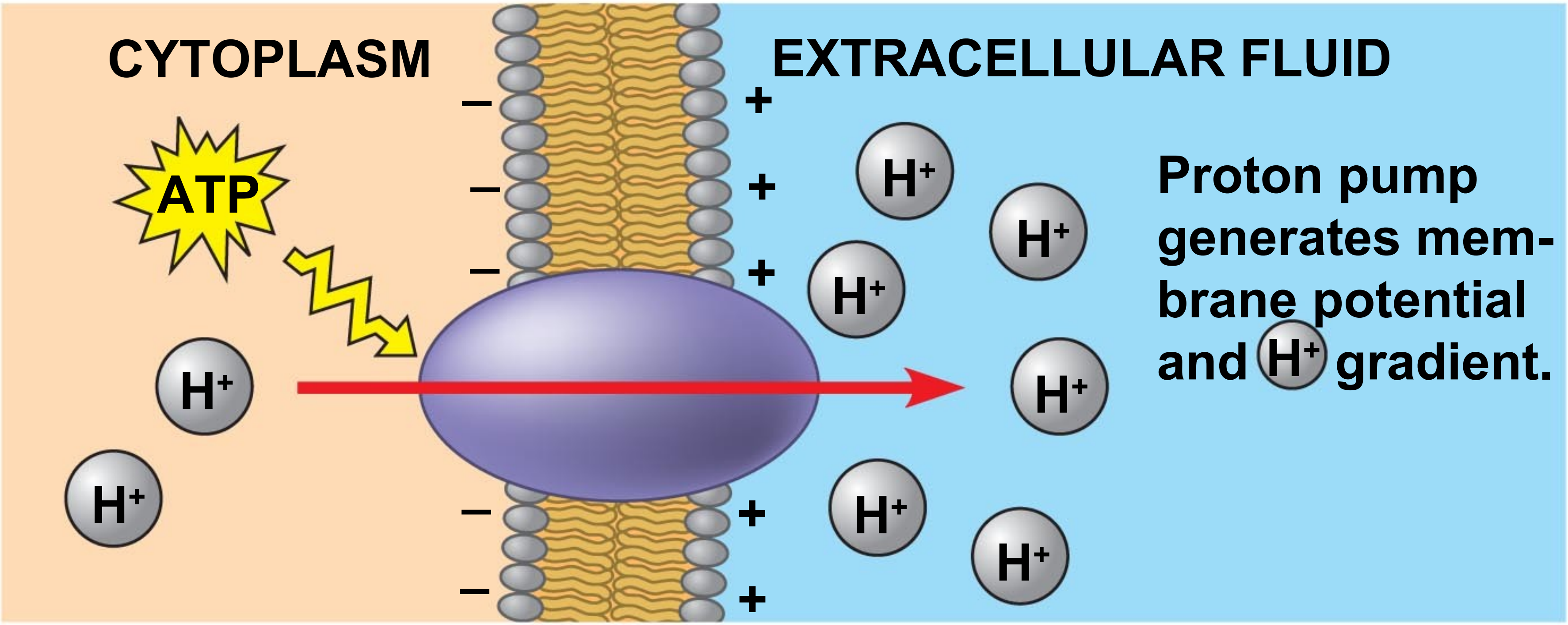
2.5 mm

Transport occurs by short-distance diffusion or active transport and by long-distance bulk flow

- Transport begins with the absorption of resources by plant cells.
- The movement of substances into and out of cells is regulated by selectively permeable membrane.
- Diffusion across a membrane is passive transport. The pumping of solutes across a membrane is active transport and requires energy.
- Most solutes pass through **transport proteins** embedded in the cell membrane.

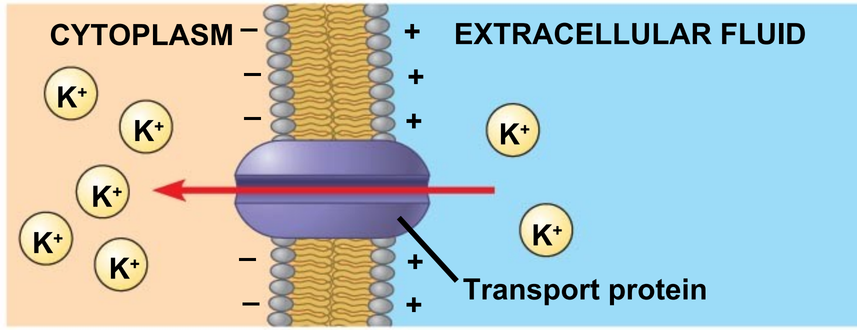
- The most important transport protein for active transport is the **proton pump**.
- Proton pumps in plant cells create a hydrogen ion gradient that is a form of potential energy that can be harnessed to do work.
- They contribute to a voltage known as a **membrane potential**.

Proton pumps provide energy for solute transport

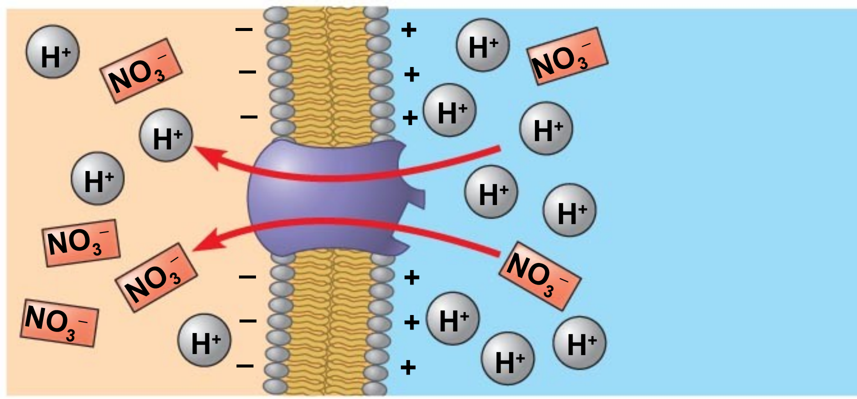


- ***Plant cells use energy stored in the proton gradient and membrane potential to drive the transport of many different solutes***.
- The “coat-tail” effect of cotransport is also responsible for the uptake of the sugar sucrose by plant cells.

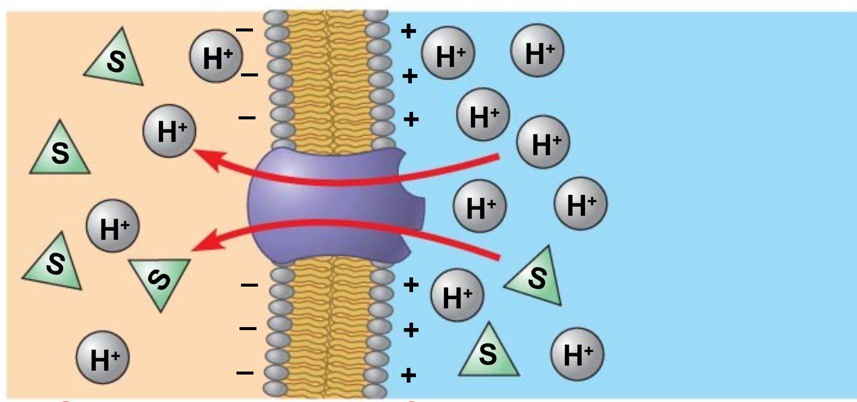
Solute transport in plant cells



(a) Membrane potential and cation uptake

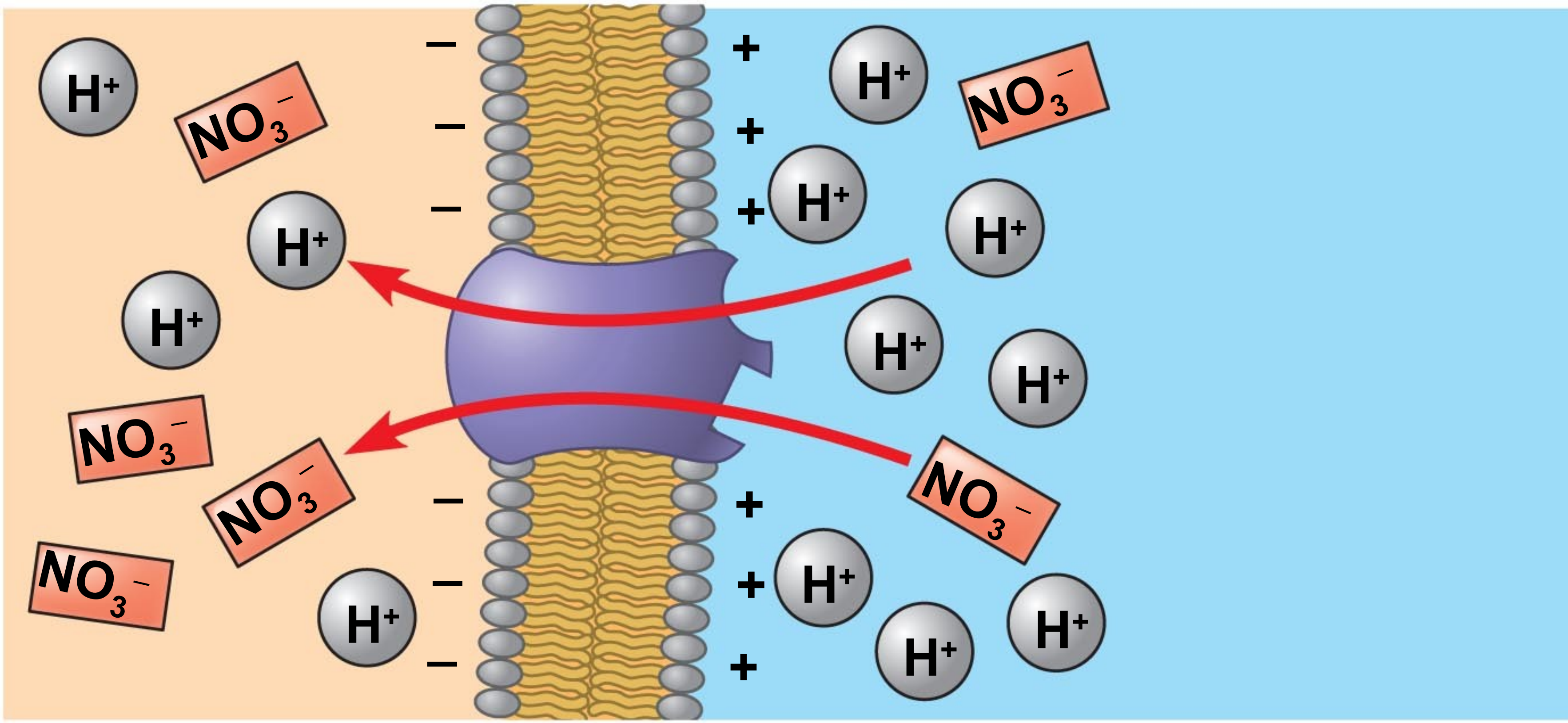


(b) Cotransport of an anion with H^+



(c) Cotransport of a neutral solute with H^+

Cotransport - a transport protein couples the diffusion of one solute to the active transport of another.



Cotransport of an anion with H^+

Diffusion of Water = Osmosis

- To survive, plants must balance water uptake and loss.
- **Osmosis** determines the net uptake or water loss by a cell and is affected by solute concentration and pressure.

- **Water potential** is a measurement that combines the effects of solute concentration and pressure.
- Water potential determines the direction of water movement.
- Water flows from regions of higher water potential to regions of lower water potential.

- Water potential is measured in units of pressure called **megapascals (MPa)**
- **Water potential = 0 MPa for pure water at sea level and room temperature**.

How Solutes and Pressure Affect Water Potential

- *Both pressure and solute concentration affect water potential*.
- The **solute potential** of a solution is proportional to the number of dissolved molecules (solutes).
- Solute potential is also called **osmotic potential.**
- **Remember: More solute** means **less water.**

- **Pressure potential** is the physical pressure on a solution.
- **Turgor pressure** is the pressure exerted by the plasma membrane against the cell wall, and the cell wall against the protoplast.

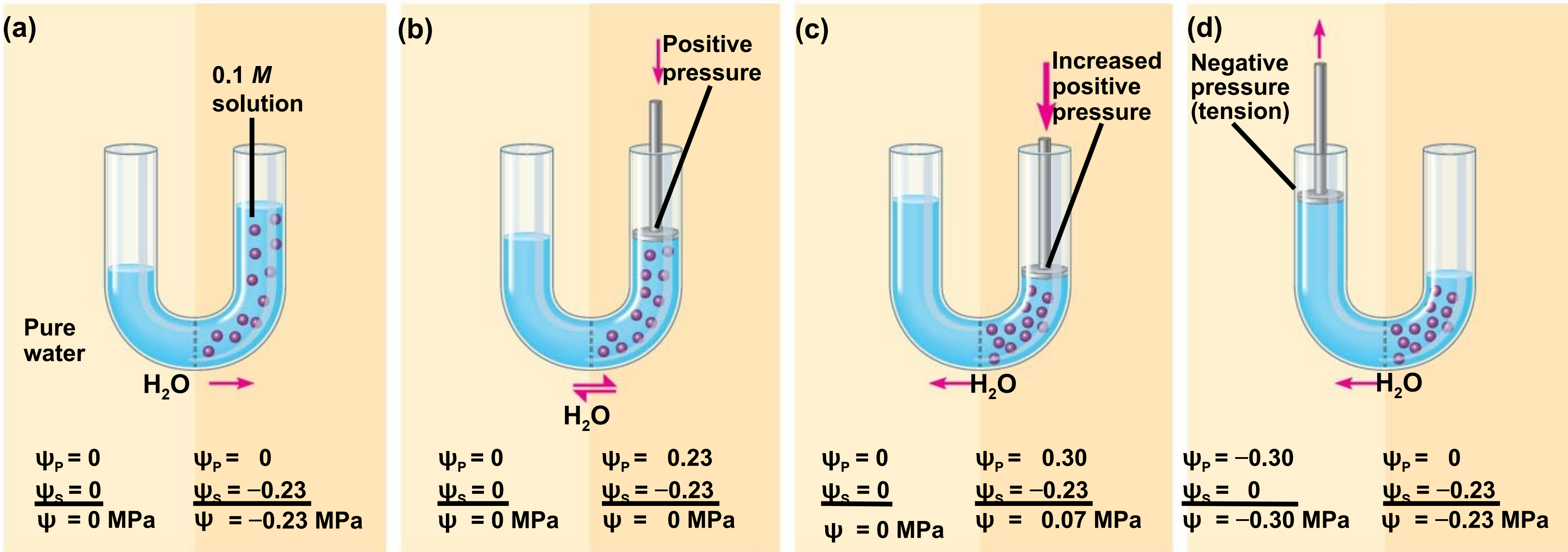
Measuring Water Potential

- Consider a U-shaped tube where the two arms are separated by a membrane permeable only to water.

- Water moves in the direction from higher water potential to lower water potential.

H --> L

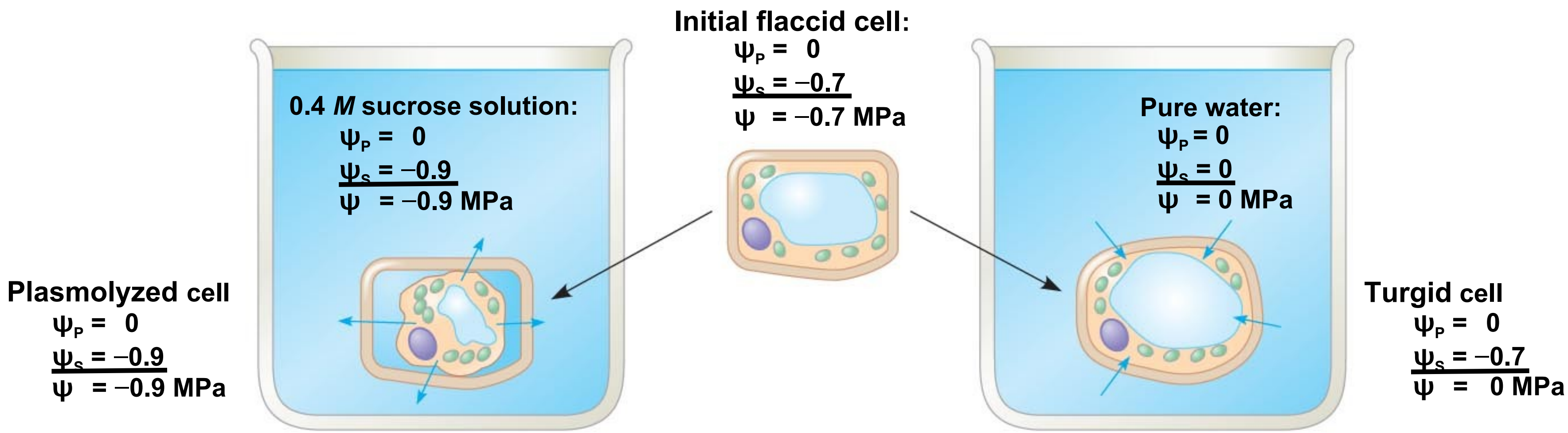
Water potential and water movement.



Addition of Solutes reduces water potential.

- ***Physical pressure increases water potential.***
- ***Negative pressure decreases water potential.***
- Water potential affects uptake and loss of water by plant cells.
- If a **flaccid** cell from an isotonic solution is placed in an environment with a higher solute concentration, the cell will lose water and undergo **plasmolysis.**
- If the same flaccid cell is placed in a solution with a lower solute concentration, the cell will gain water and become **turgid.**

Water relations in plant cells



(a) Initial conditions: cellular ψ > environmental ψ

(b) Initial conditions: cellular ψ < environmental ψ

- Turgor loss in plants causes **wilting**, which can be reversed when the plant is watered.
- **Aquaporins** are transport proteins in the cell membrane that allow the passage of water.
- The rate of water movement is likely regulated by phosphorylation of the aquaporin proteins.

A wilted Impatiens plant regains its turgor when watered

Cells in wilted plant to the left - plasmolysis

Cells in plant below - turgor.

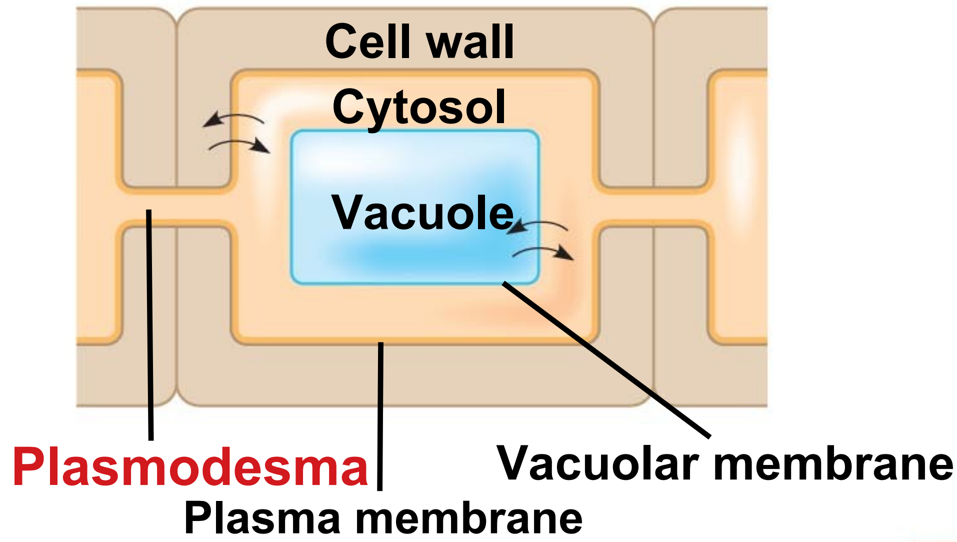
Three Major Pathways of Transport

- Transport is also regulated by the compartmental structure of plant cells.
- The plasma membrane directly controls the traffic of molecules into and out of the protoplast.
- The plasma membrane is a barrier between two major compartments, the cell wall and the cytosol.

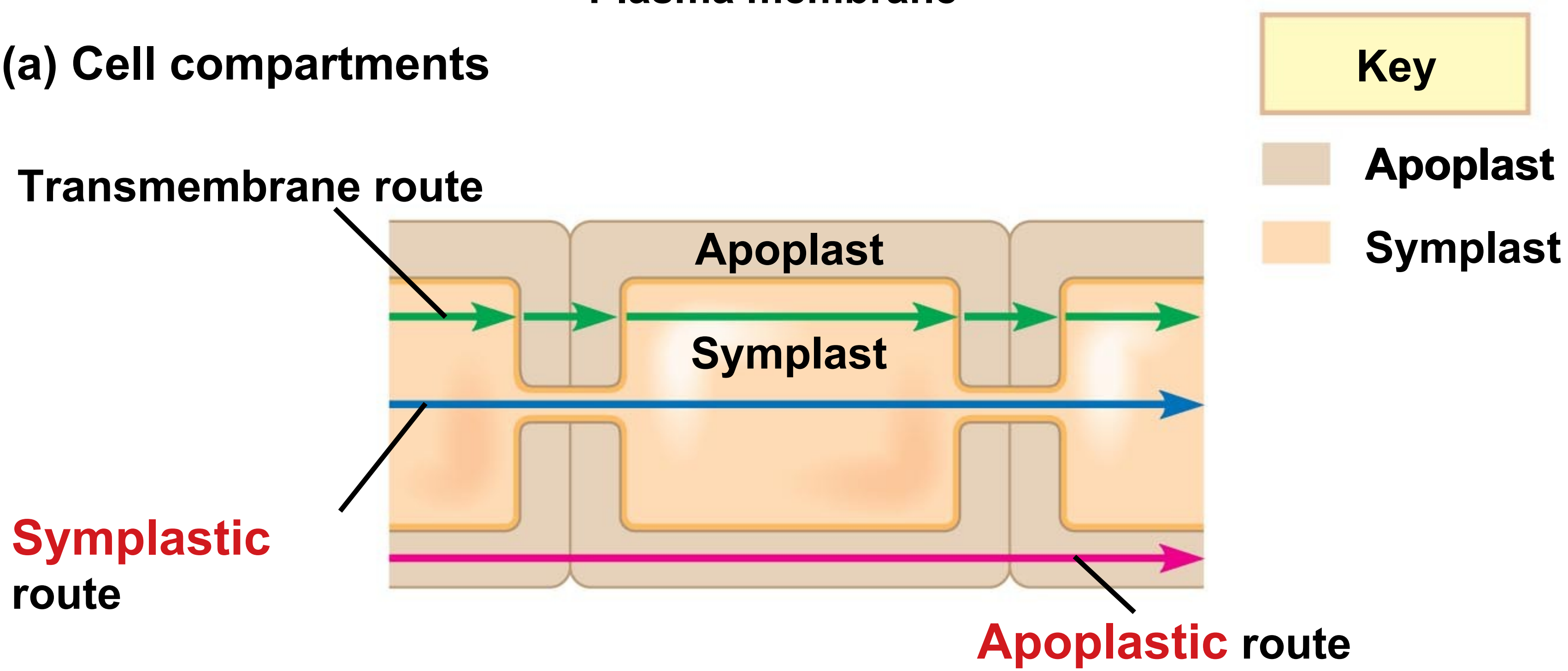
- The third major compartment in most mature plant cells is the central vacuole, a large organelle that occupies as much as 90% or more of the protoplast’s volume.
- The vacuolar membrane = tonoplast - regulates transport between the cytosol and the vacuole.

- In most plant tissues, the cell wall and cytosol are continuous from cell to cell.
- The cytoplasmic continuum is called the **symplast**.
- The cytoplasm of neighboring cells is connected by channels = **plasmodesmata**.
- The **apoplast** is the continuum of cell walls and extracellular spaces.

Short Distance Transport



(a) Cell compartments



(b) Transport routes between cells

Water and Mineral Short Distance Transport

- Water and minerals can travel through a plant by three routes:
  - Transmembrane route: out of one cell, across a cell wall, and into another cell
  - Symplastic route: via the continuum of cytosol
  - Apoplastic route: via the cell walls and extracellular spaces

Bulk Flow in Long-Distance Transport -Vessels Xylem and Phloem

- Efficient long distance transport of fluid requires **bulk flow**, the movement of a fluid driven by pressure.
- Water and solutes move together through tracheids and vessel elements of xylem, and sieve-tube elements of phloem.
- Efficient movement is possible because mature tracheids and vessel elements have no cytoplasm, and sieve-tube elements have few organelles in their cytoplasm.

Absorption of Water and Minerals by Root Cells

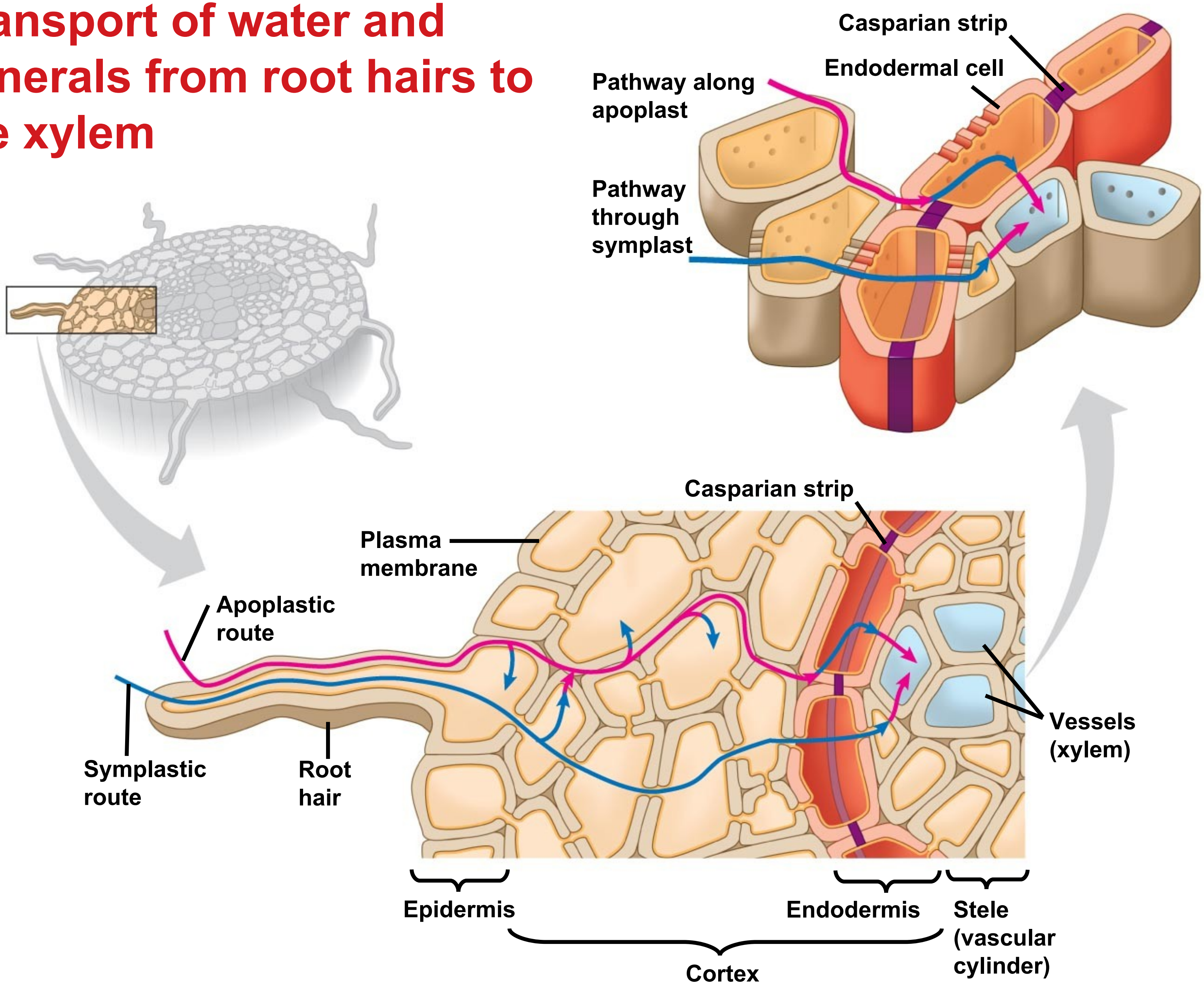
- Most water and mineral absorption occurs near root tips, where the epidermis is permeable to water and root hairs are located.
- Root hairs account for much of the surface area of roots.
- After soil solution enters the roots, the extensive surface area of cortical cell membranes enhances uptake of water and selected minerals.

Transport of Water and Minerals into the Xylem

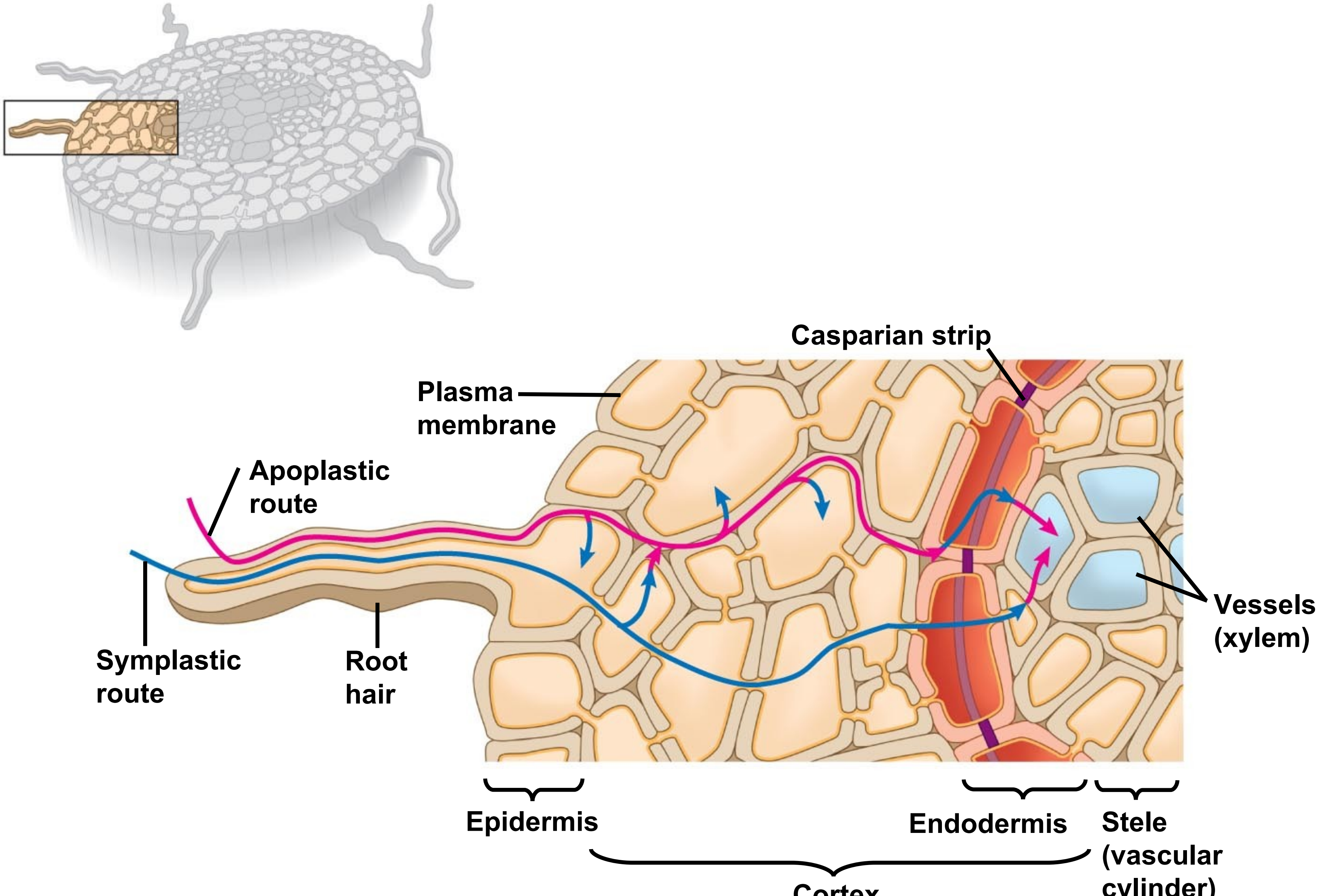
- The **endodermis** is the innermost layer of cells in the root cortex.
- It surrounds the vascular cylinder and is the last checkpoint for selective passage of minerals from the cortex into the vascular tissue.

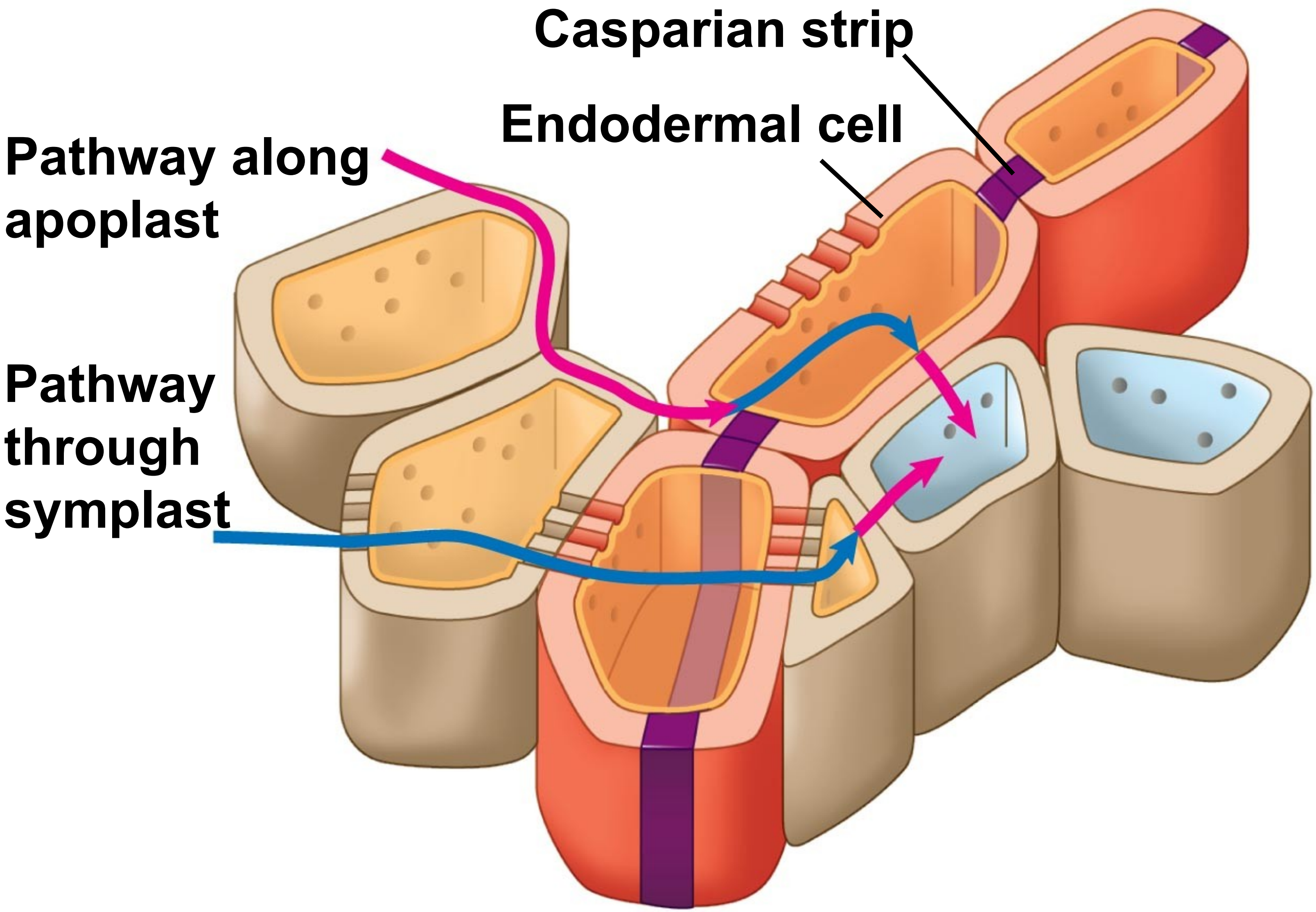
- Water can cross the cortex via the symplast or apoplast.
- The waxy **Casparian strip** of the endodermal wall blocks apoplastic transfer of minerals from the cortex to the vascular cylinder.

Transport of water and minerals from root hairs to the xylem



Transport of water and minerals from root hairs to the xylem




Casparian strip
Endodermal cell
Pathway along apoplast
Pathway through symplast

Bulk Flow Driven by Negative Pressure in the Xylem

- Plants lose a large volume of water from **transpiration**, the evaporation of water from a plant's surface. This creates a negative pressure at the stomata opening (where water was lost).

- Water is replaced by the bulk flow of water and minerals, called **xylem sap**, from the roots to the stems and leaves.

Pushing Xylem Sap: Root Pressure

- At night, when stomatas are closed, transpiration is very low. Root cells continue pumping mineral ions into the xylem of the vascular cylinder, lowering the water potential.
- Water flows in from the root cortex, generating **root pressure**.
- Root pressure sometimes results in **guttation**, the exudation of water droplets on tips or edges of leaves … usually in small plants.
- Positive root pressure is relatively weak and is a minor mechanism of xylem bulk flow.

Guttation

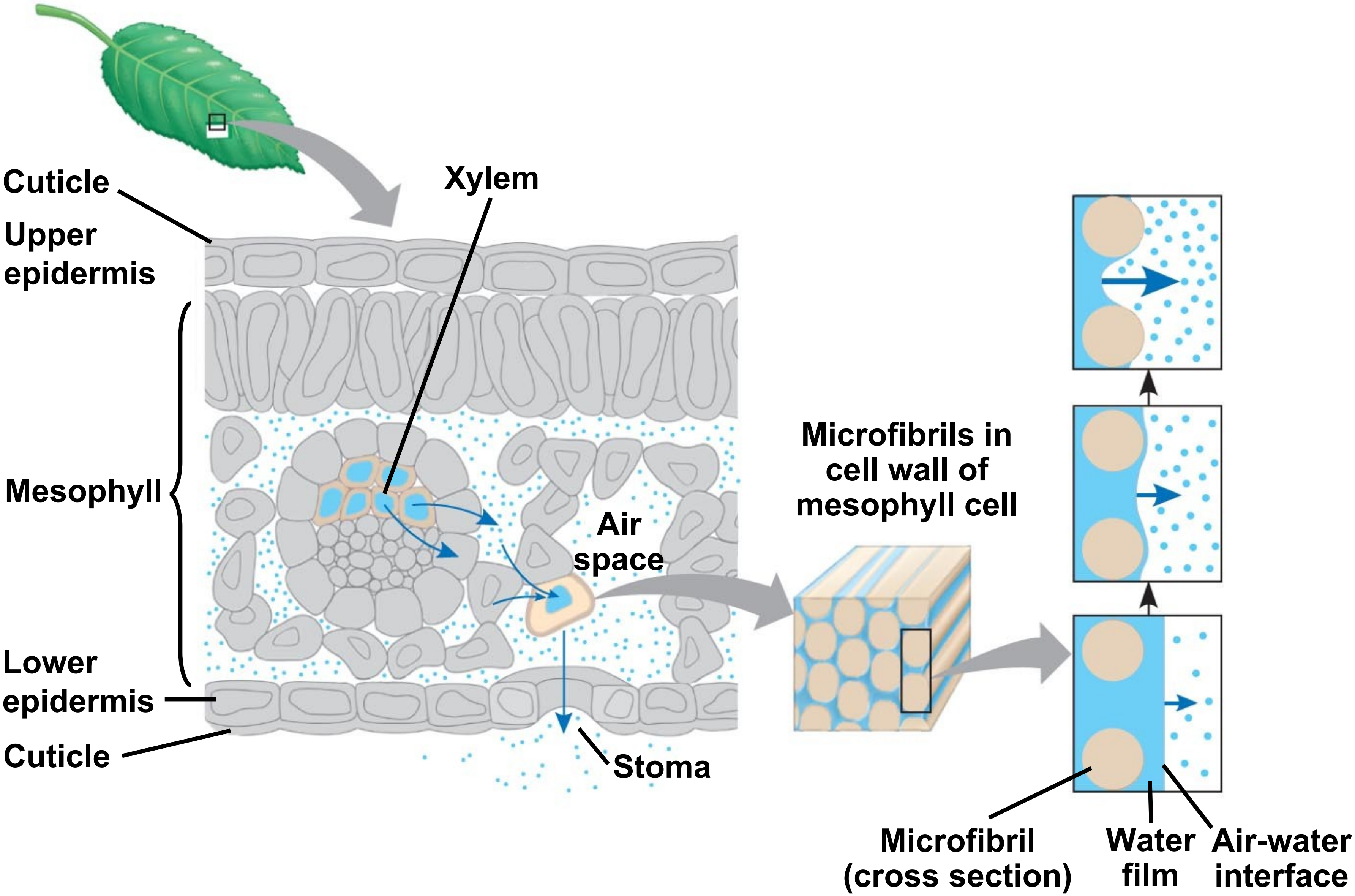
Pulling Xylem Sap: The Transpiration-Cohesion-Tension Mechanism

- ***Water is pulled upward by negative pressure in the xylem***

Transpiration Pull:

- Water vapor in the airspaces of a leaf diffuses down its water potential gradient and exits the leaf via stomata. (This creates a low - a negative pressure).
- Transpiration produces negative pressure (tension) in the leaf, which exerts a pulling force on water in the xylem, pulling water into the leaf. (H --> L).

Generation of transpiration pull

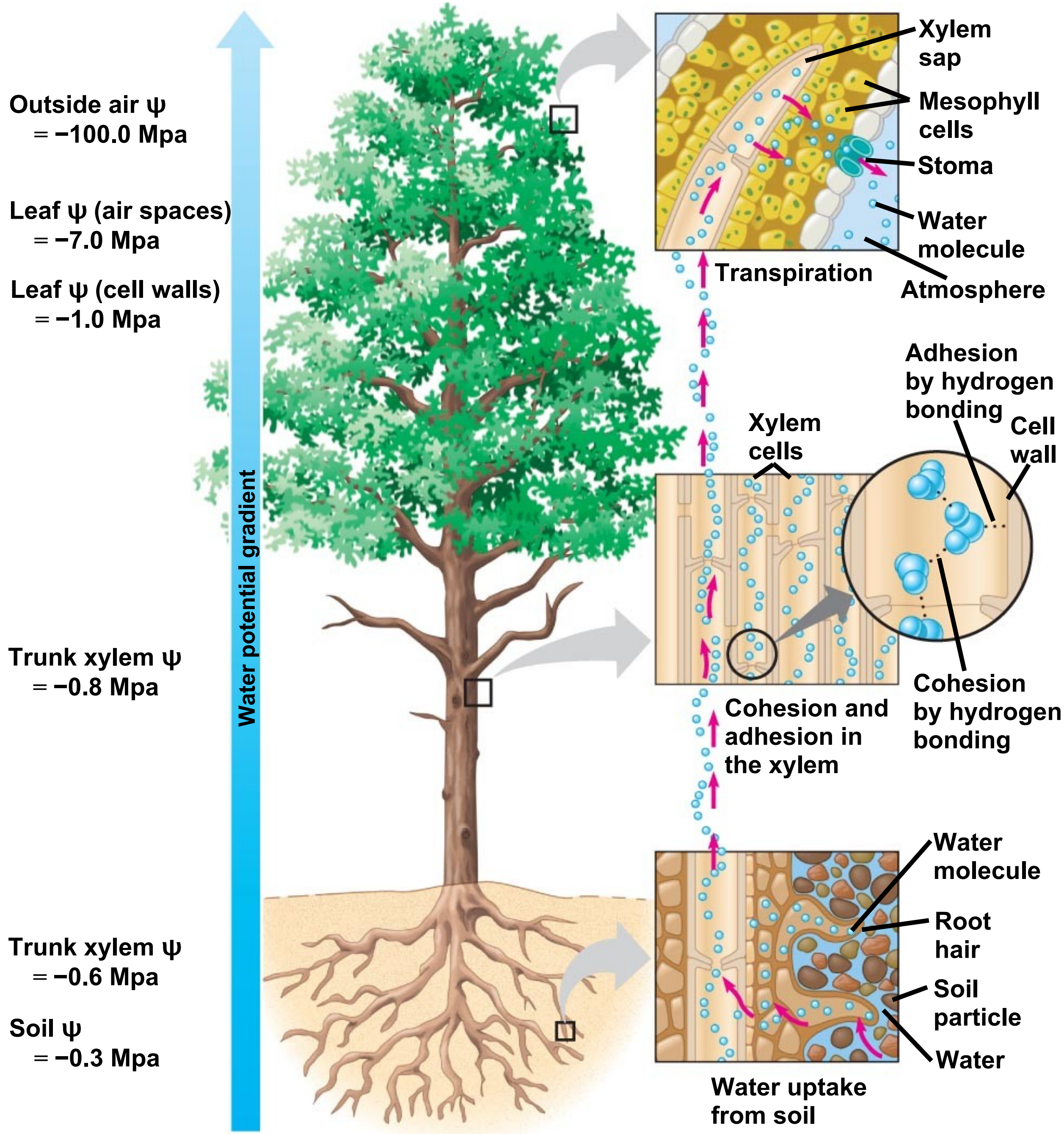


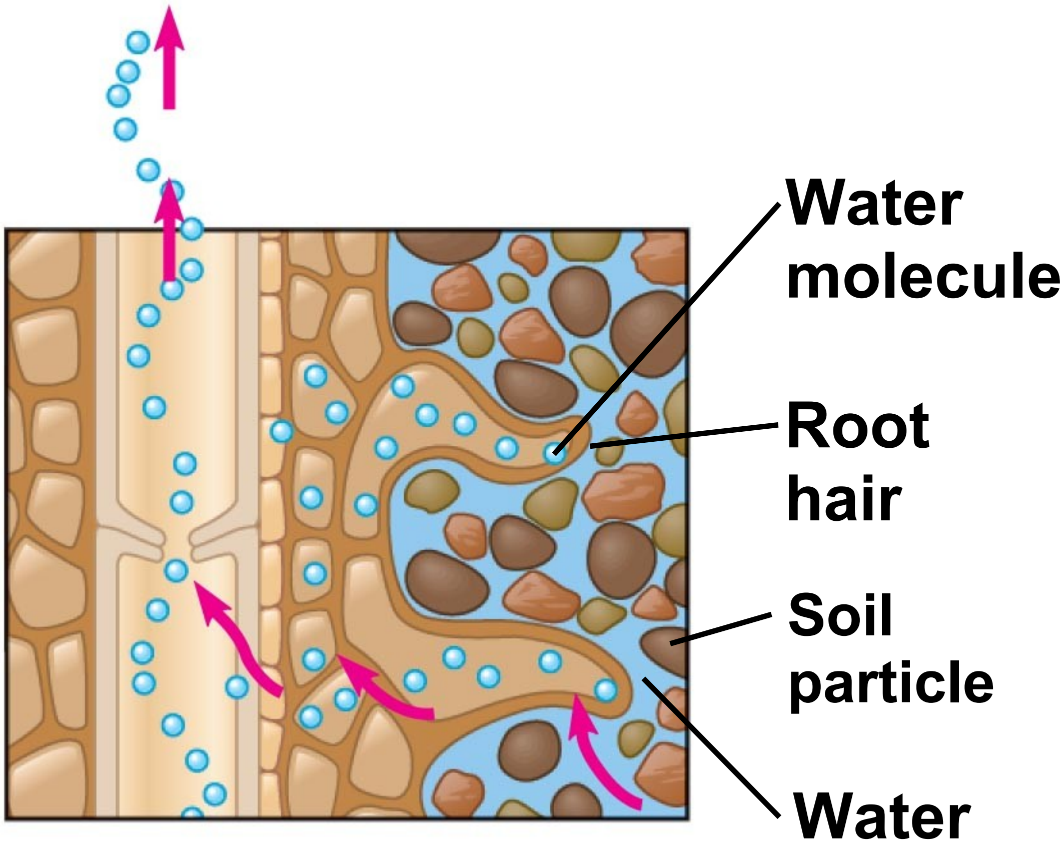
Cohesion and Adhesion in the Ascent of Xylem Sap

- The transpirational pull on xylem sap is transmitted all the way from the leaves to the root tips and even into the soil solution.
- Transpirational pull is facilitated by cohesion of water molecules to each other (so water column rises unbroken) and adhesion of water molecules to the xylem vascular tissue.

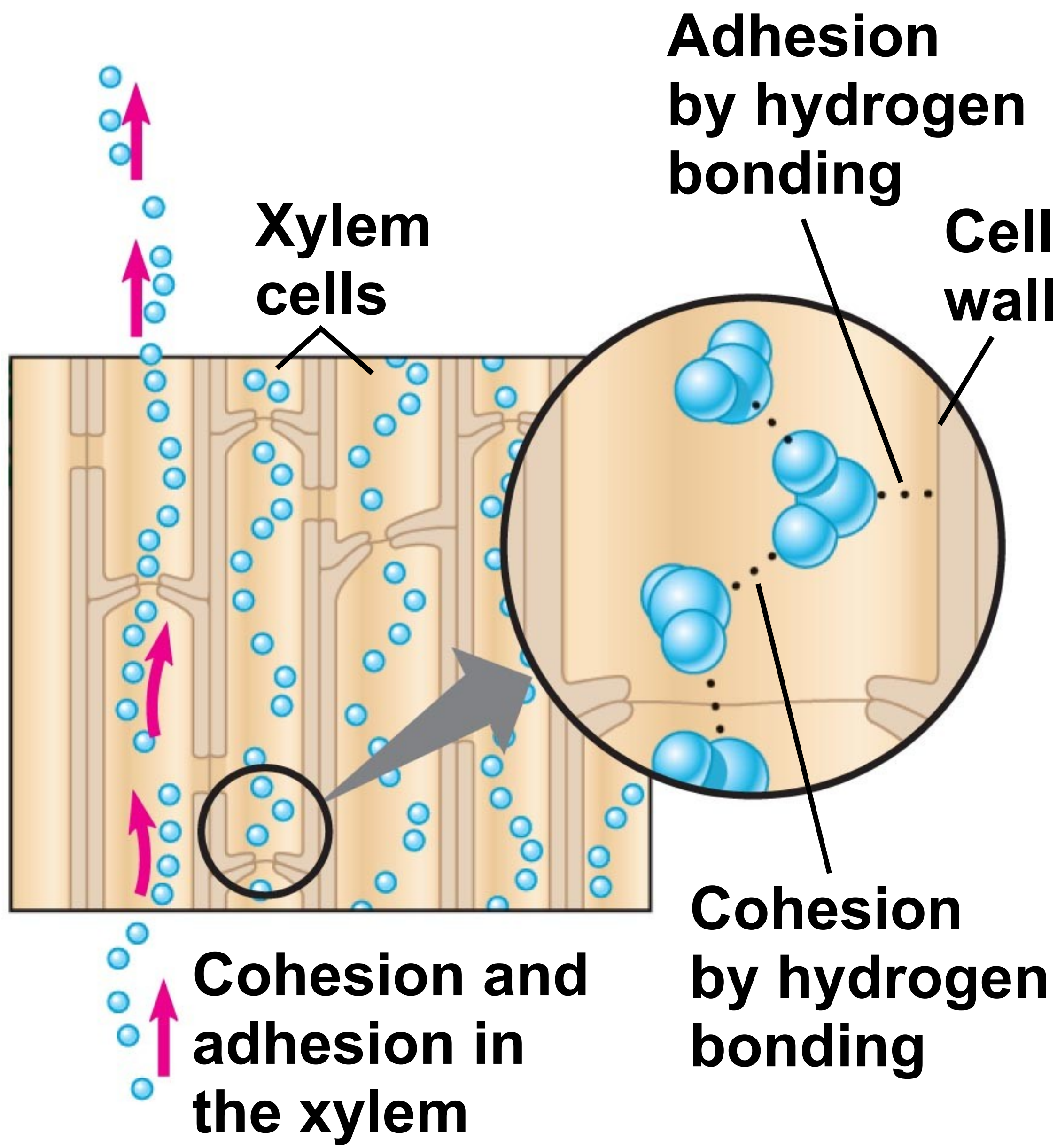
- Drought stress or freezing can cause cavitation, the formation of a water vapor pocket by a break in the chain of water molecules. This can be fatal to the plant.

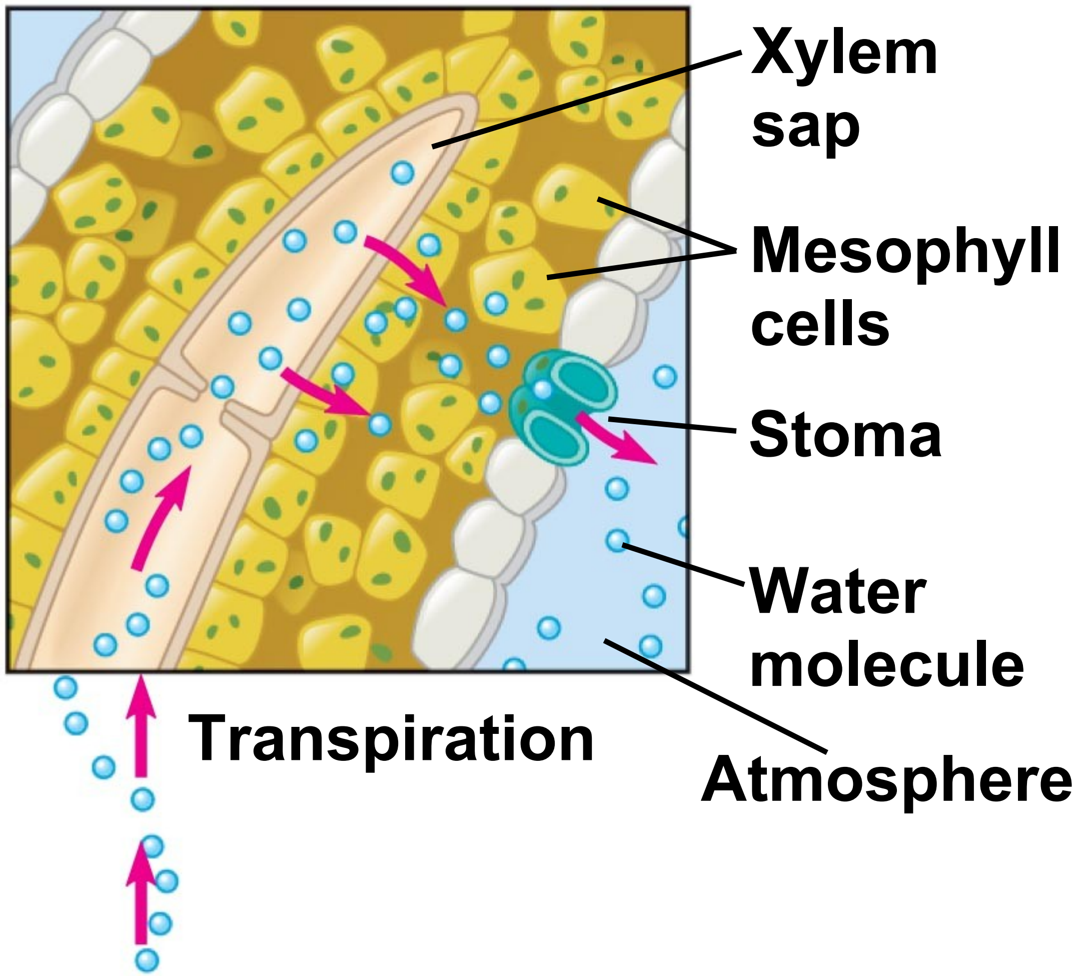
Ascent of xylem sap





Water uptake
from soil


Adhesion by hydrogen bonding
Cell wall
Xylem cells
Cohesion by hydrogen bonding
Cohesion and adhesion in the xylem


Xylem sap
Mesophyll cells
Stoma
Water molecule
Atmosphere
Transpiration

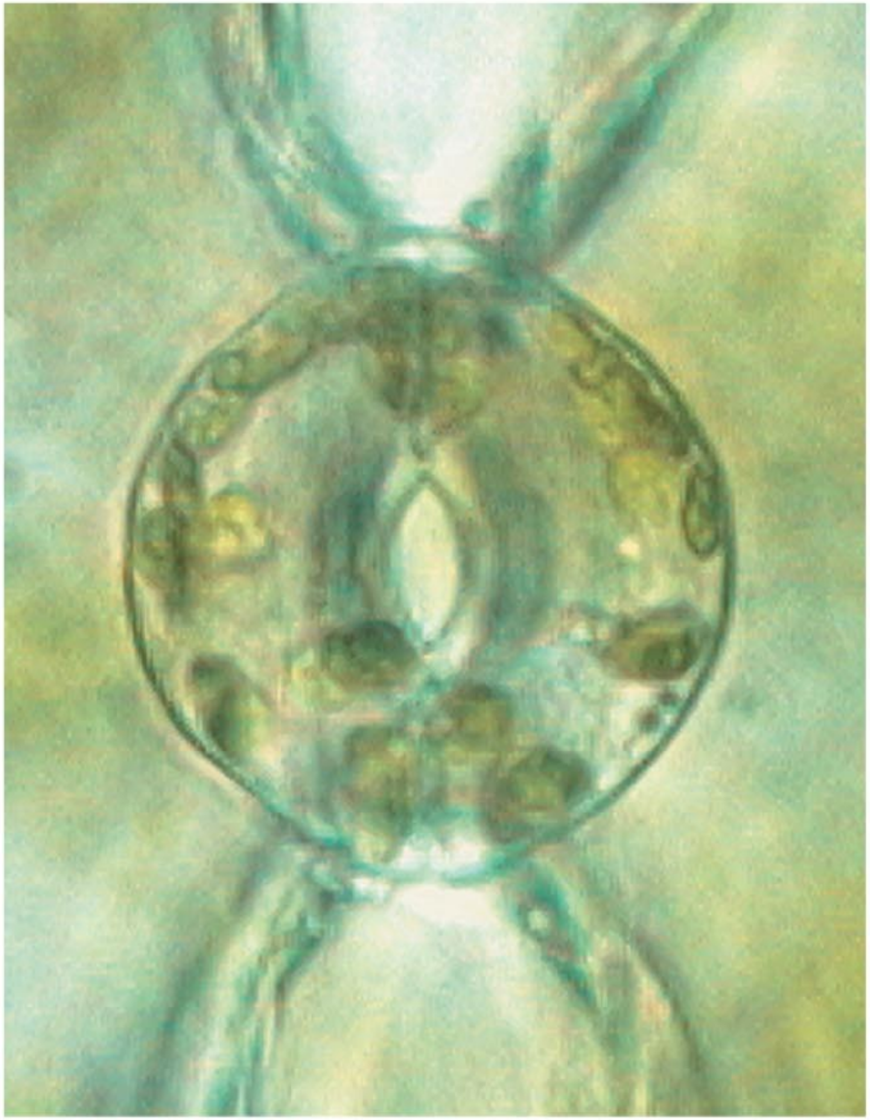
Xylem Sap Ascent by Bulk Flow: *A Review*

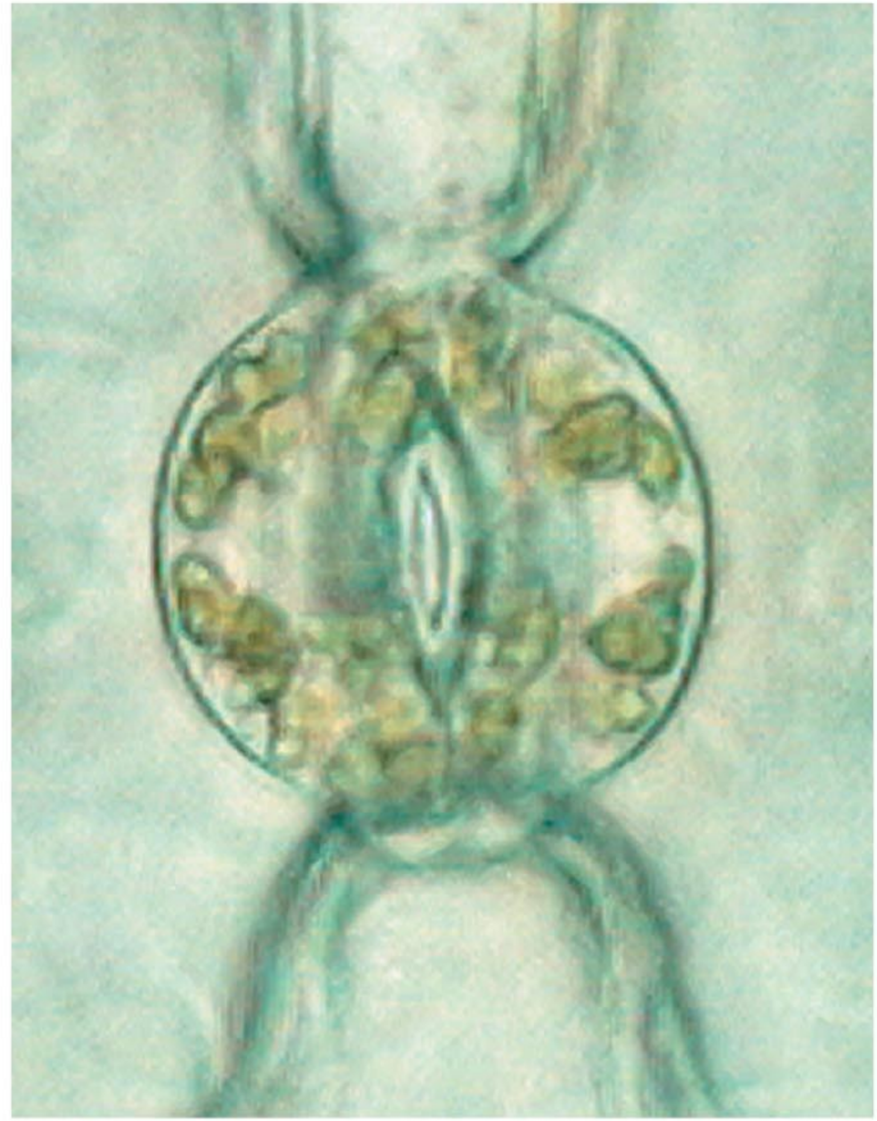
- The movement of xylem sap against gravity is maintained by the ***transpiration-cohesion-tension mechanism***.
- Transpiration lowers water potential in leaves, and this generates negative pressure (tension) that pulls water up through the xylem.
- There is no energy cost to bulk flow of xylem sap.

Stomata help regulate the rate of transpiration

- Leaves generally have broad surface areas and high surface-to-volume ratios.
- These characteristics increase photosynthesis and increase water loss through stomata.
- About 95% of the water a plant loses escapes through stomata.
- Each stoma is flanked by a pair of **guard cells**, which control the diameter of the stoma by changing shape.

An open stoma (left) and closed stoma (right)

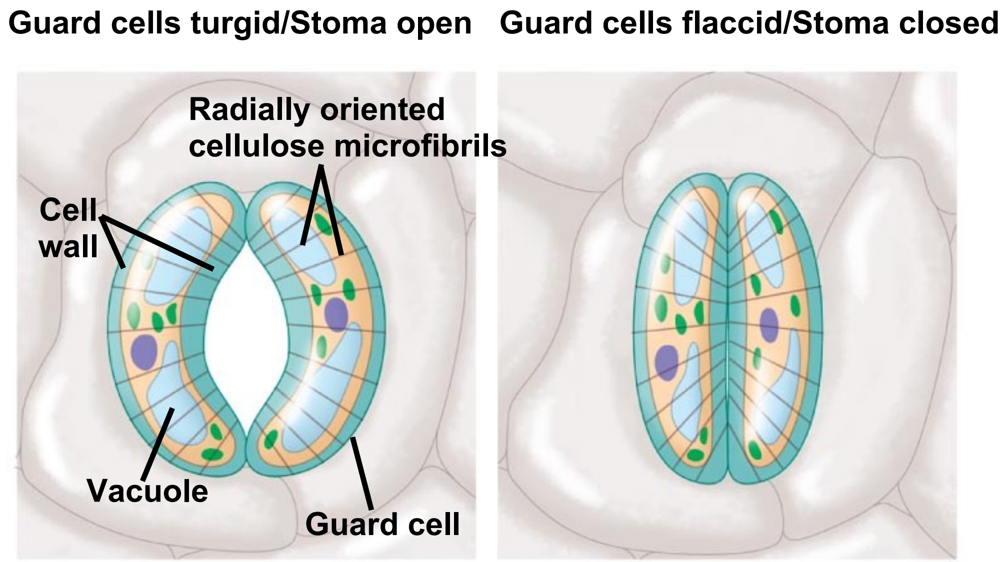




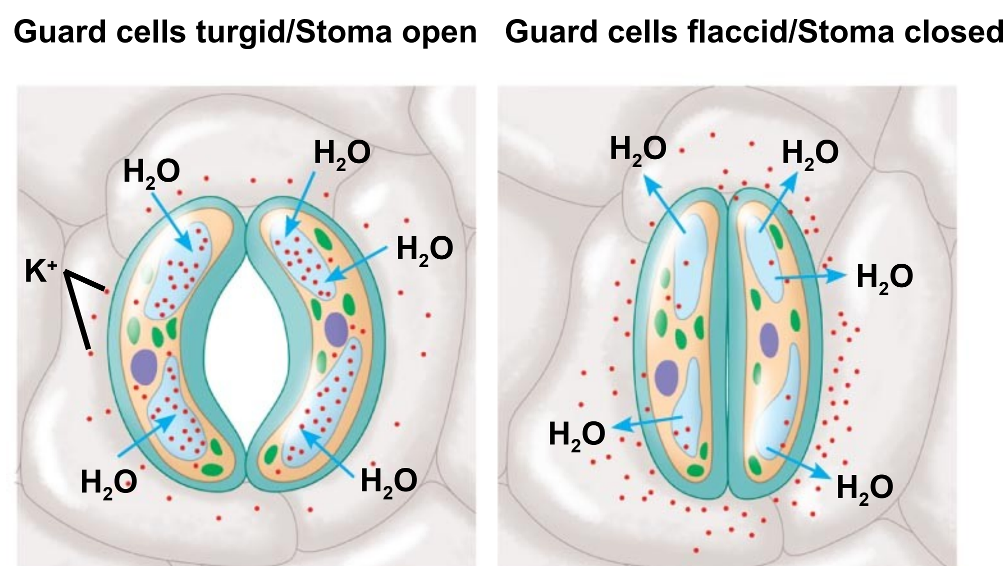
Mechanisms of Stomatal Opening and Closing

- Changes in ***turgor pressure*** open and close stomata.
- These result primarily from the reversible uptake and loss of ***potassium ions*** by the guard cells.

Stomatal Openings



(a) Changes in guard cell shape and stomatal opening and closing



(b) Role of potassium ions in stomatal opening and closing

Stimuli for Stomatal Opening and Closing

- Generally, ***stomata open during the day and close at night to minimize water loss***.
- Stomatal opening at dawn is triggered by:
  - light,
  - CO_2 depletion, and
  - an internal "clock" in guard cells.
- All eukaryotic organisms have internal clocks; **circadian rhythms** are 24-hour cycles.

Effects of Transpiration on Wilting and Leaf Temperature

- Plants lose a large amount of water by transpiration.
- If the lost water is not replaced by sufficient transport of water, the plant will lose water and wilt.
- Transpiration also results in evaporative cooling, which can lower the temperature of a leaf and prevent denaturation of various enzymes involved in photosynthesis and other metabolic processes.

Adaptations That Reduce Evaporative Water Loss

- **Xerophytes** are plants adapted to arid climates.
- They have leaf modifications that reduce the rate of transpiration.
- *Some plants use a specialized form of photosynthesis* called crassulacean acid metabolism ***CAM*** where *stomatal gas exchange occurs at night*.

Xerophytic - Desert Plants

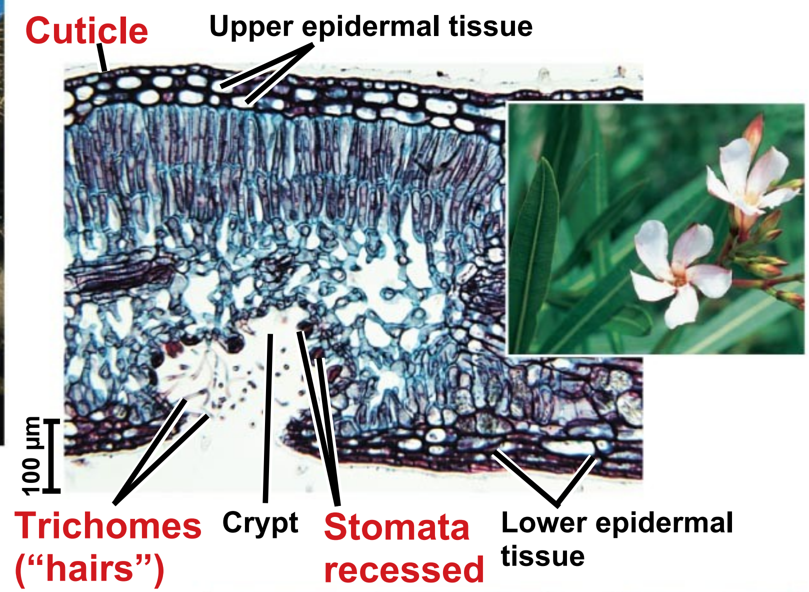
Adaptations

Ocotillo - leafless

Ocotillo after heavy rain

Ocotillo leaves after a heavy rain

Oleander leaf cross section and flowers



Old man cactus

Sugars are transported from leaves and other sources to sites of use or storage

- The products of photosynthesis are transported through phloem by the process of **translocation**.

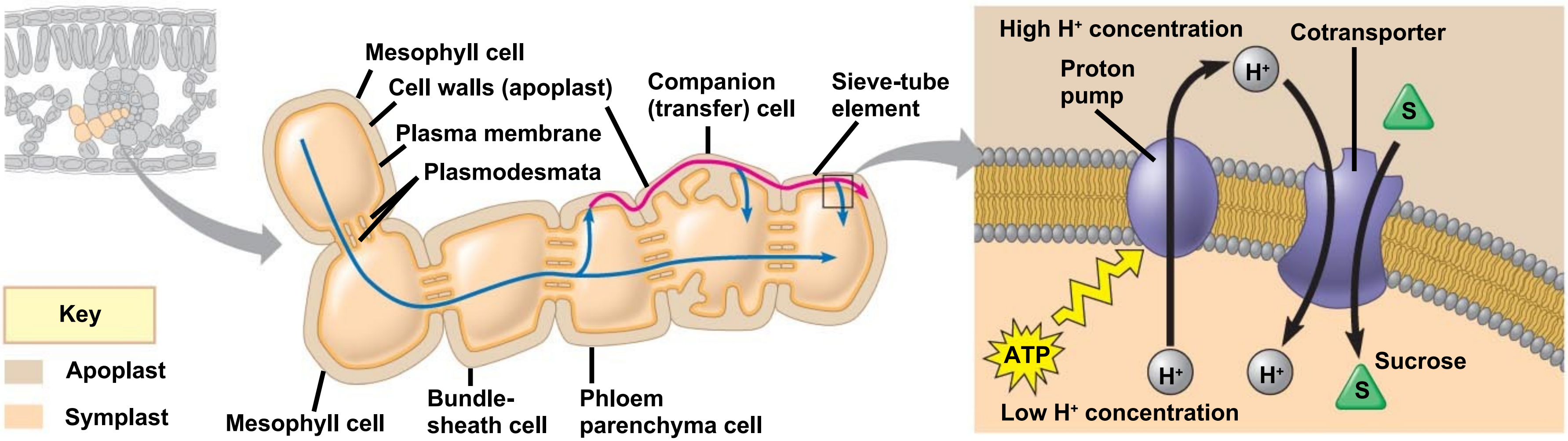
Movement from Sugar Sources to Sugar Sinks

- **Phloem sap** is an aqueous solution that is high in sucrose = disaccharide.
- It travels from a sugar source to a sugar sink:
  Source to sink
- A **sugar source** is an organ that is a net producer of sugar, such as mature leaves.
- A **sugar sink** is an organ that is a net consumer or storer of sugar, such as a tuber or bulb.
- *A storage organ can be both a sugar sink in summer and sugar source in winter*.

Phloem: Translocaton: source to sink

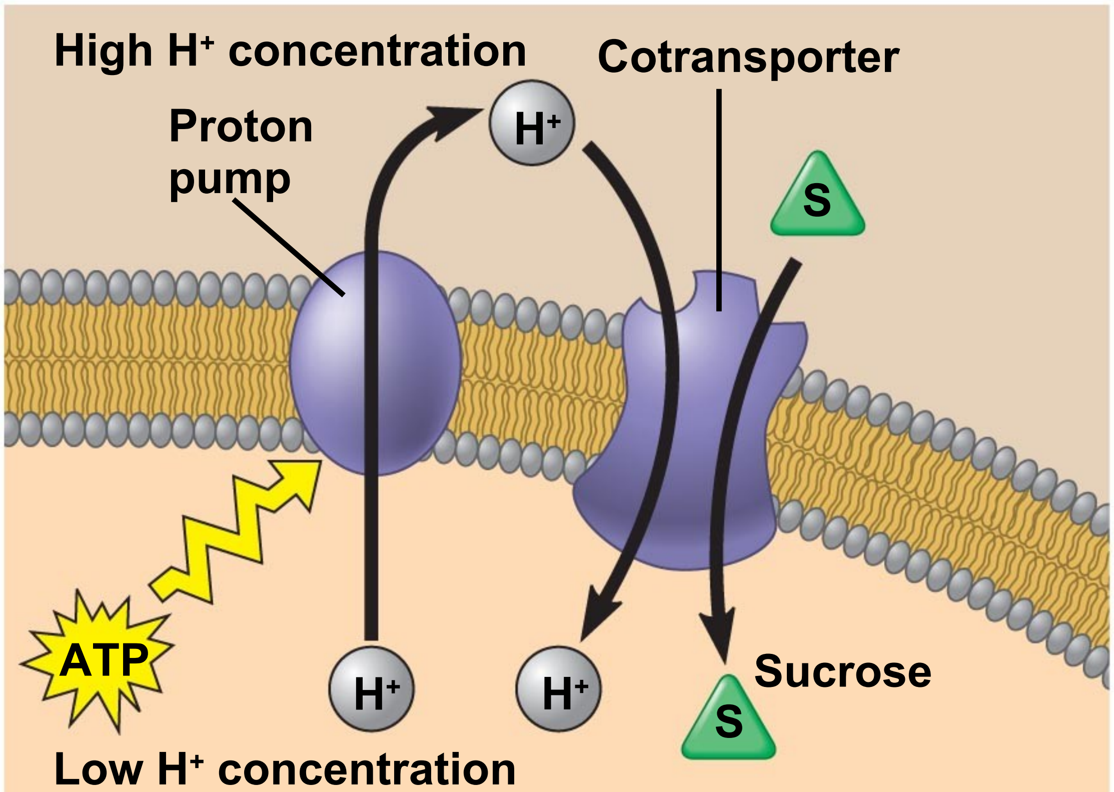
- Sugar must be loaded into sieve-tube elements before being exposed to sinks.
- Depending on the species, sugar may move by symplastic or both symplastic and apoplastic pathways.
- **Transfer cells** are modified companion cells that enhance solute movement between the apoplast and symplast.

Loading of sucrose into phloem
proton pump -- Cotransport of Sucrose



- In many plants, phloem loading requires active transport.
- **Proton pumping and cotransport of sucrose and H^+** enable the cells to accumulate sucrose.
- At the sink, *sugar molecules are transported from the phloem to sink tissues* and are *followed by water*.

Loading of sucrose into phloem: Cotransport



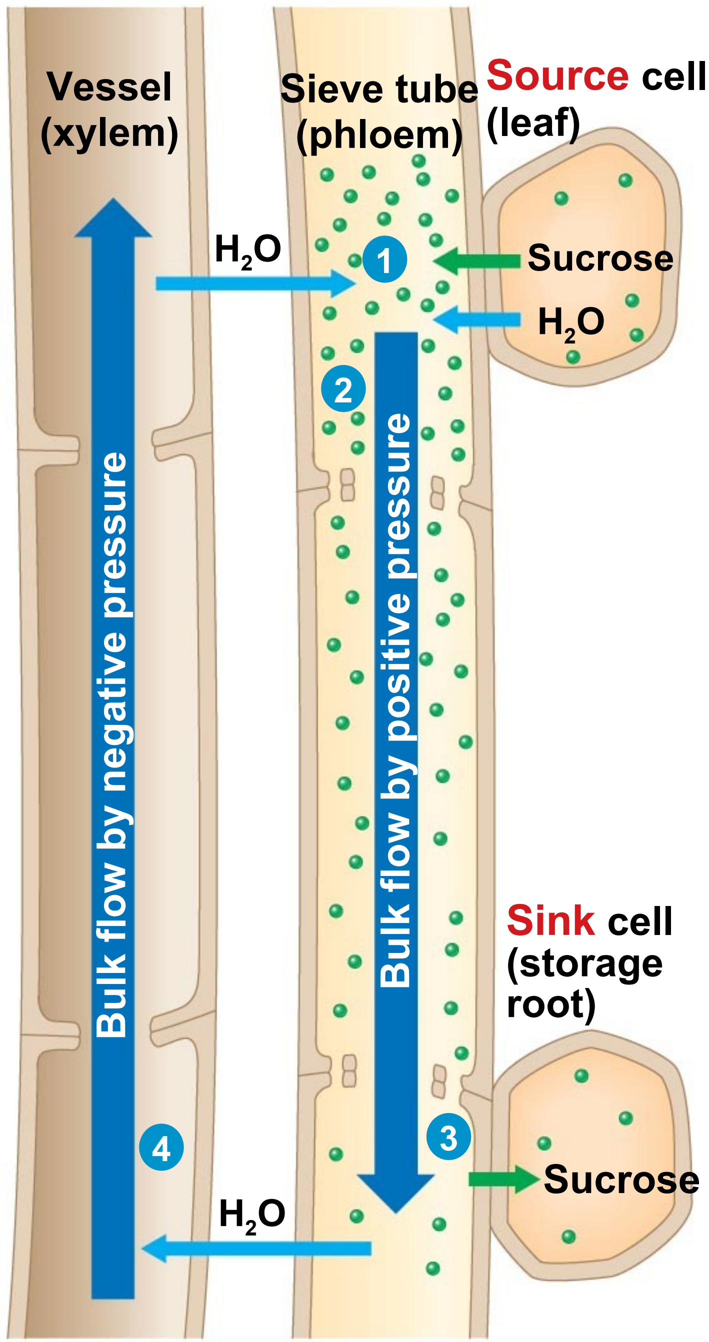
Bulk Flow by Positive Pressure: The Mechanism of Translocation in Angiosperms

- In studying angiosperms, researchers have concluded that sap moves through a sieve tube by bulk flow driven by positive pressure.

- The ***pressure flow hypothesis explains why phloem sap always flows from source to sink***.

Bulk flow by positive pressure.

Pressure Flow in a sieve tube

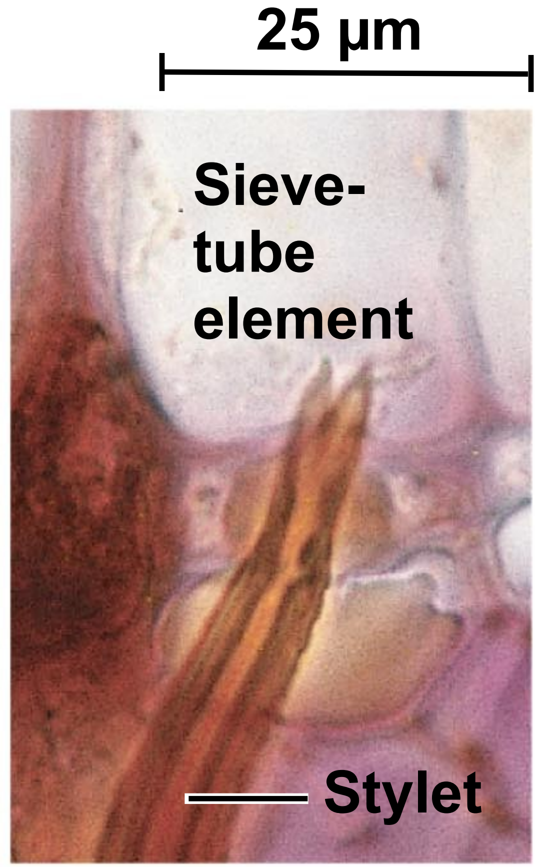


1. Loading of sugar
2. Uptake of water
3. Unloading of sugar
4. Water recycled

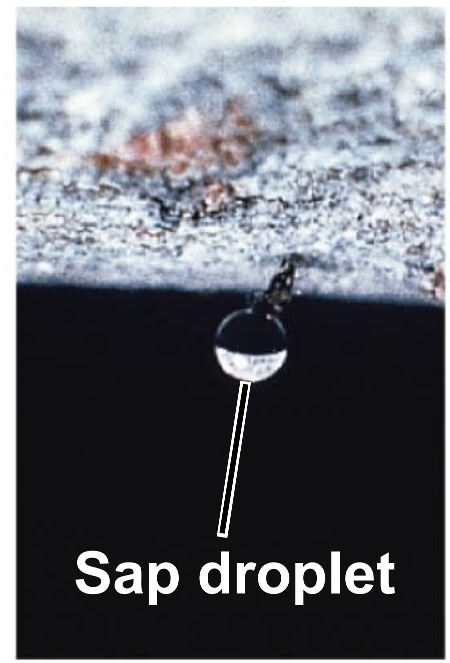
EXPERIMENT

Does phloem sap contain more sugar near sources than sinks?

Aphid feeding



Stylet in sieve-tube element



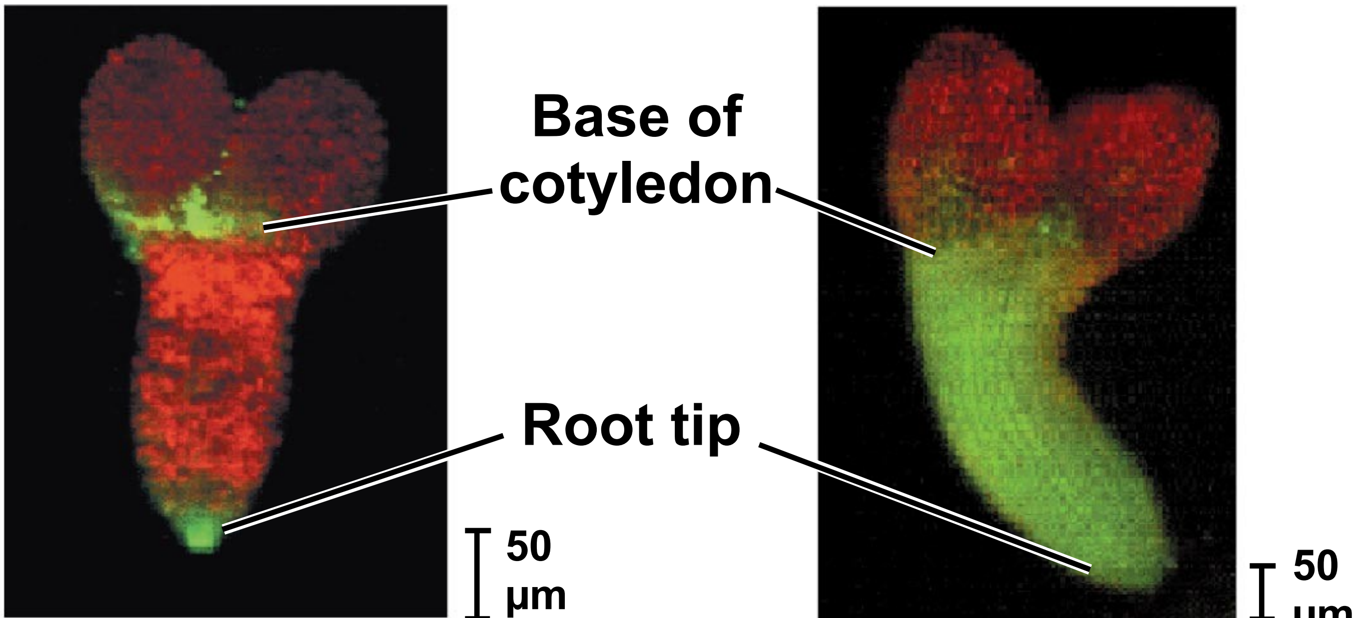
Separated stylet exuding sap

The Symplast is highly dynamic - Plasmodesmata - Continuously Changing Structures

- The symplast is a living tissue and is responsible for dynamic changes in plant transport processes.
- Plasmodesmata can change in permeability in response to turgor pressure, cytoplasmic calcium levels, or cytoplasmic pH.
- Plant viruses can cause plasmodesmata to dilate
- Mutations that change communication within the symplast can lead to changes in development.

Question: Do alterations in symplastic communication affect plant development?

EXPERIMENT Results

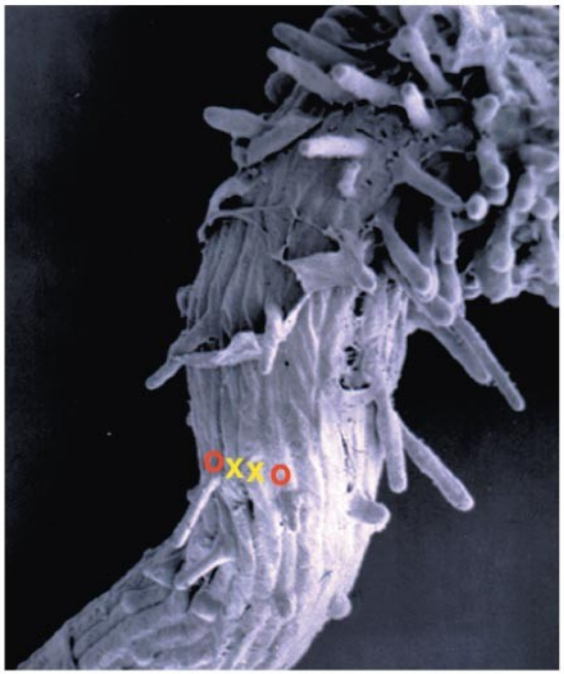


Wild-type embryo

Mutant embryo

Question: Do alterations in symplastic communication affect plant development?

Experiment RESULTS



50 µm

Wild-type seedling root tip

50 µm

Mutant seedling root tip

Electrical Signaling in the Phloem

- The phloem allows for rapid electrical communication between widely separated organs.
- Phloem is a "superhighway" for systemic transport of macromolecules and viruses.
- **Systemic** communication helps integrate functions of the whole plant.

You should now be able to:

1. Describe how proton pumps function in transport of materials across membranes.
2. Define the following terms: osmosis, water potential, flaccid, turgor pressure, turgid.
3. Explain how aquaporins affect the rate of water transport across membranes.
4. Describe three routes available for short-distance transport in plants.

5. Relate structure to function in sieve-tube cells, vessel cells, and tracheid cells.

6. Explain how the endodermis functions as a selective barrier between the root cortex and vascular cylinder.

7. Define and explain guttation.

8. Explain this statement: “The ascent of xylem sap is ultimately solar powered.”

9. Describe the role of stomata and discuss factors that might affect their density and behavior.

10. Trace the path of phloem sap from sugar source to sugar sink; describe sugar loading and unloading.